Interim Report

From Fragmented Journeys to Unified Experiences:

Investigating Multifunctional Mobile App Design Through a Fitness Case Study

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Project Overview

The purpose of this study is to examine if multifunctional mobile design may facilitate more cohesive and effective user journeys and how fragmented digital journeys impact user experience in modern mobile applications. As a practical case study, the project focuses on a mobile fitness application because this field frequently necessitates that users interact with several separate tools—such as timers, workout trackers, note-taking apps, and progress visualization—to accomplish a single real-world task.

Modern mobile usage is increasingly characterised by brief, sporadic interactions and frequent context switching. Despite the fact that many apps are well-designed when used in isolation, users usually need to transition between them to achieve their goals, which results in fragmented workflows and an increase in cognitive load. This research was motivated by the fact that such fragmentation places the burden of integration on the user rather than the system, potentially jeopardising usability, efficiency, and long-term engagement.

The project takes a user journey-centred approach, focusing on the continuity of experience across tasks and features rather than discrete interface-level evaluation. The study looks at how effectively a system supports progression, orientation, and job flow over time rather than whether individual screens are useful. This viewpoint is in line with current ideas in service design and human-computer interaction, where experience quality is thought to result from linked interaction sequences rather than isolated touchpoints.

The project looks at the opposing strategies of multi-purpose platforms and single-purpose applications in order to underpin this inquiry. Background study on super app models has been done as part of the project's advancement thus far, including a comparison of the ecosystems

of WeChat and Meta. A cognitive walkthrough that has been carried out as part of the project work to evaluate how well these platforms support coherent user journeys in practice has been informed by this research. These results are being used to inform future design choices.

The project's practical component is designing and developing a prototype of a multipurpose workout application aimed at casual gym patrons. The prototype intends to combine popular fitness-related technologies into a single mobile application, including timers, note-taking, gym attendance tracking, workout planning, and progress visualization. The goal is to establish a controlled environment where journey continuity and cognitive strain may be investigated, rather than to fully reproduce the features of commercial fitness programs.

Early phases of the project's iterative development were devoted to requirements collection, research, and design exploration. The next steps will be the prototype's implementation, peer usability testing, and critical evaluation of the results. By combining design research, practical development, and user-centred assessment, the project seeks to shed light on how multifunctional mobile applications might be developed to reduce fragmentation and promote more unified user experiences.

Progress Made

With reference to the initial project concept and the project's changing research focus, this section provides an overview of the work completed during the first half of the project. The majority of the work that has been done thus far has focused on requirements definition, research, analytical evaluation, and early design exploration. When taken as a whole, these efforts have created a solid conceptual and practical basis for the creation and assessment of prototypes in the project's second half.

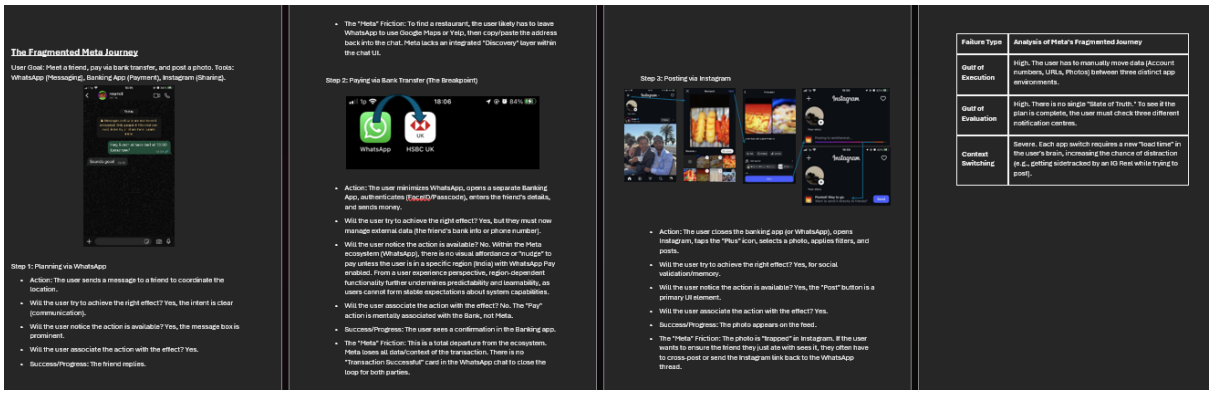
2.1 Research and Analytical Progress

Establishing the project's theoretical and analytical foundation has advanced significantly. A thorough analysis of the literature has been

finished, looking at important topics such service design, multifunctional platform architectures, fragmented digital journeys, mobile usability, and human–computer interaction. The project's focus was refined as a result of this research, moving away from interface-level usability heuristics and toward a more comprehensive, journey-oriented viewpoint that takes into account task flow, cognitive load, and continuity throughout mobile engagements.

To embed theoretical ideas in current practice, applied study of real-world platforms has been conducted in addition to analysing available literature. WeChat and Meta's ecosystem were chosen as contrasted case studies for background study on multipurpose "super app" models because of their similar functional scope but different integration and user experience design strategies.

An early cognitive walkthrough was conducted as part of the project's analytical work, building on this study. To investigate how users move through related jobs inside each ecosystem, a uniform, multi-step scenario was implemented to both platforms. The walkthrough concentrated on feedback systems, task continuity, navigation consistency, and the cognitive effects of context shift. WeChat facilitated more continuous user journeys inside a single interaction environment, but Meta's ecosystem frequently required transfers between numerous distinct programs to complete related tasks, according to the report.



The Fragmented Meta Journey

User Goal: Meet a friend, pay via bank transfer, and post a photo. Tools: WhatsApp (Messaging), Banking App (Payment), Instagram (Sharing).

Step 1: Planning via WhatsApp

- Action: The user sends a message to a friend to coordinate the location.
- Will the user try to achieve the right effect? Yes, the intent is clear (communication).
- Will the user notice the action is available? Yes, the message box is prominent.
- Will the user associate the action with the effect? Yes.
- Success/Progress: The friend replies.

- The "Meta" Friction: To find a restaurant, the user likely has to leave WhatsApp to use Google Maps or Yelp, then copy/paste the address back into the chat. Meta lacks an integrated "Discovery" layer within the chat UI.

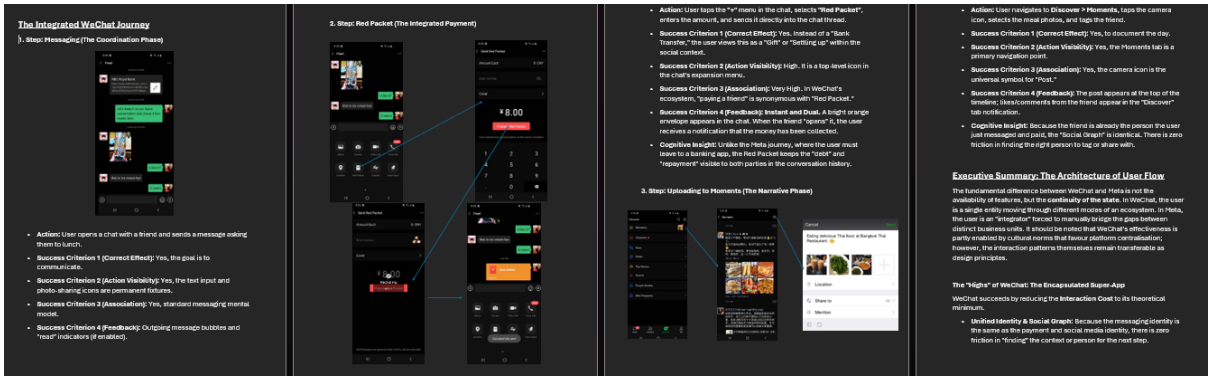
Step 2: Paying via Bank Transfer (The Breakpoint)

- Action: The user minimizes WhatsApp, opens a separate Banking App, authenticates (FaceID/Passcode), enters the friend's details, and sends money.
- Will the user try to achieve the right effect? Yes, but they must now manage external data (the friend's bank info or phone number).
- Will the user notice the action is available? No. Within the Meta ecosystem (WhatsApp), there is no visual affordance or "trudge" to pay unless the user is in a specific region (India) with WhatsApp Pay enabled. From a user experience perspective, region-dependent functionality further undermines predictability and learnability, as users cannot form stable expectations about system capabilities.
- Will the user associate the action with the effect? No. The "Pay" action is mentally associated with the Bank, not Meta.
- Success/Progress: The user sees a confirmation in the Banking app.
- The "Meta" Friction: This is a total departure from the ecosystem. Meta loses all data/context of the transaction. There is no "Transaction Successful" card in the WhatsApp chat to close the loop for both parties.

Step 3: Posting via Instagram

- Action: The user closes the banking app (or WhatsApp), opens Instagram, taps the "Plus" icon, selects a photo, applies filters, and posts.
- Will the user try to achieve the right effect? Yes, for social validation/memory.
- Will the user notice the action is available? Yes, the "Post" button is a primary UI element.
- Will the user associate the action with the effect? Yes.
- Success/Progress: The photo appears on the feed.
- The "Meta" Friction: The photo is "trapped" in Instagram. If the user wants to ensure the friend they just ate with sees it, they often have to cross-post or send the Instagram link back to the WhatsApp thread.

Failure Type	Analysis of Meta's Fragmented Journey
Gulf of Execution	High. The user has to manually move data (Account numbers, URLs, Photos) between three distinct app environments.
Gulf of Evaluation	High. There is no single "State of Truth." to see if the plan is complete, the user must check three different notification centres.
Context Switching	Severe. Each app switch requires a new "load time" in the user's brain, increasing the chance of distraction (e.g., getting sidetracked by an IG Reel while trying to post).



These results immediately influenced the project's design direction and offered verifiable proof of route fragmentation in modern mobile ecosystems, highlighting the significance of examining a multifunctional strategy.

2.2 Requirements Gathering and Feature Definition

The project advanced to requirements collection for the fitness application prototype, utilizing insights from the super app analysis and literature review. The intended user demographic was determined to be regular, casual gym patrons who are neither fitness specialists nor professional athletes. This group was selected because they relied on simple, time-efficient tools and frequently used multiple apps during exercises.

Based on shared user objectives and observed fragmentation within existing fitness app ecosystems, core functional requirements were determined. These prerequisites consist of:

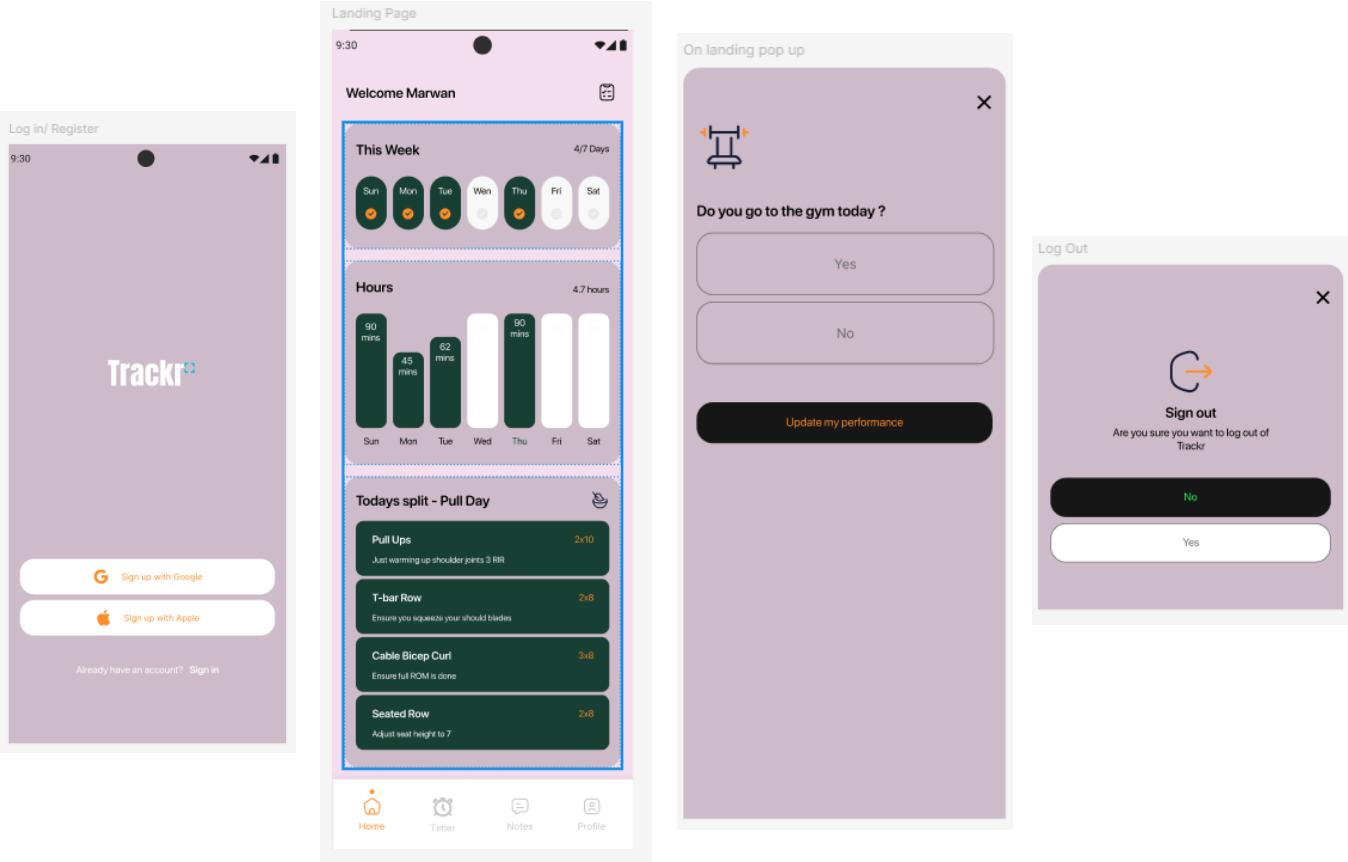
- Tracking gym attendance over time with basic visualisation
- Making and overseeing exercise splits
- Developing and overseeing exercise splits
- Timing rest periods and workouts
- Monitoring basic progress metrics

Requirements were established with a focus on journey continuity rather than feature depth. In order to minimize the need for numerous tools and minimize context switching, the goal is to allow a coherent sequence of actions—planning, performing, timing, recording, and analysing

workouts—within a single application. These specifications will direct further design choices and serve as the foundation for a usability assessment.

2.3 Early Design Exploration

Research findings and functional needs are being translated into initial interface concepts in early design development. To investigate feature grouping, navigation structure, and general application flow, preliminary sketches and low-fidelity wireframes were created. At this stage, information architecture and user journeys have taken priority over intricate visual styling in design activities.



Minimizing navigation depth, preserving uniform interaction patterns across features, and facilitating quick transition between related activities during brief interaction windows were given top priority during design exploration. The project's emphasis on lowering cognitive load and promoting fragmented attention in gym settings is reflected in these factors.

During this early design phase, potential challenges were also highlighted, such as finding a balance between feature integration and interface simplicity and avoiding user overload as capability increases. These issues will be fixed iteratively during the project's prototype and assessment phases.

Updated Project Plan & Management

This section provides an updated plan for the second half of the project, reflects on changes made to the initial plan, and describes how the project has been handled thus far. Although the first Gantt chart offered an organized framework, improvements have been made in response to new information discovered throughout the investigation and analysis stage.

3.1 Reflection on the Original Project Plan

A structured plan was created at the beginning of the project to direct progress through the stages of design, development, evaluation, and literature review. According to this approach, the prototype and evaluation would be carried out using usability heuristics as the main analytical lens. Early research and applied analysis made it evident as the project developed that this approach did not adequately portray the complexity of contemporary mobile usage.

Analysis of mobile interaction patterns and multifunctional platforms in particular revealed that many usability problems arise across fragmented user journeys spanning many applications rather than at the level of individual interfaces. This realization led to a shift in the project's focus from isolated heuristic compliance to journey continuity, cognitive load, and system-level interaction.

In order to give journey analysis, cognitive walkthroughs, and requirements specification priority, some tasks that were initially scheduled as heuristic evaluation activities were either reframed or delayed. This change was a clarification of scope that reinforced the project's alignment with current human-computer interaction research and insured that future

development effort would be more successfully grounded, rather than a loss of progress.

3.2 Project Control and Management Approach

Project progress has been managed through organised planning and iterative refinement. The original Gantt chart has been used as a baseline reference, and milestones have been regularly evaluated to assess viability and progress. Meetings with supervisors have been essential for verifying decisions, determining what has to be altered, and ensuring that the project complies with academic requirements and grading standards.

Research and design activities have been conducted using an iterative methodology, which enables requirements and early design decisions to be directly informed by findings from applied analysis and literature study. This strategy has assisted in avoiding a hasty commitment to technological implementation prior to a thorough understanding of the problem space.

Walkthrough notes, design sketches, and literature summaries are just a few of the artifacts created thus far that have been tracked and preserved through the use of version control and documentation techniques. This has made sure that direction changes are not haphazard but rather transparent and justified.

3.3 Updated Plan for the Second Half of the Project

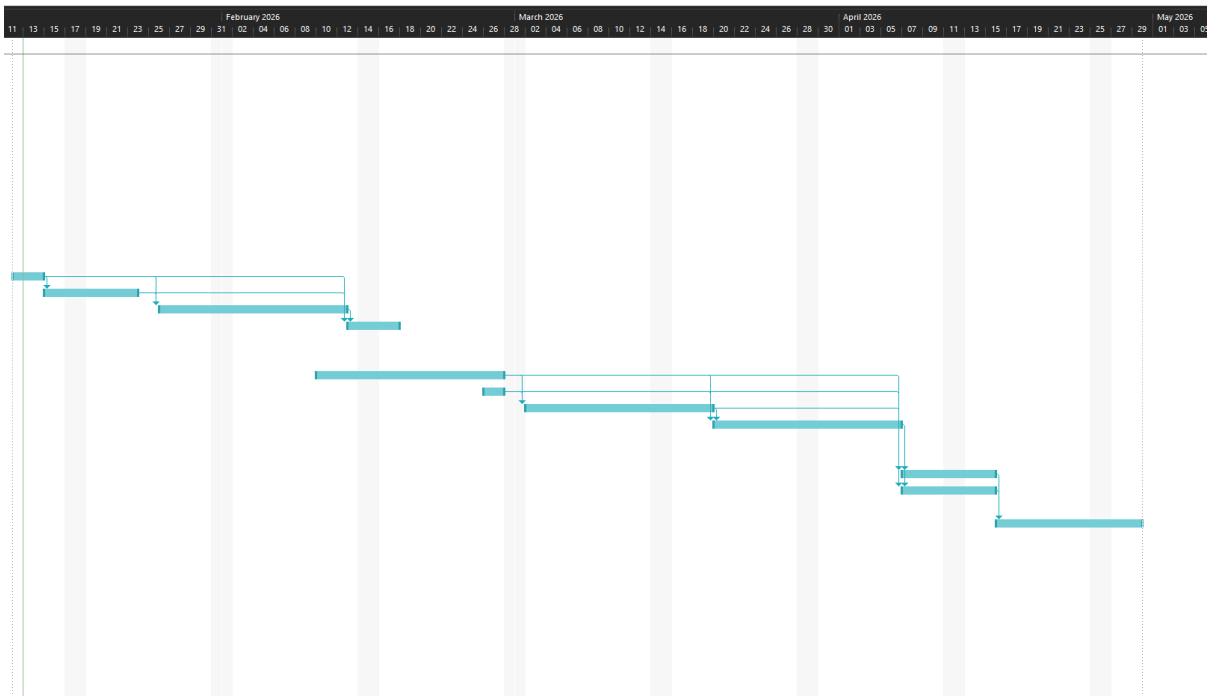
The narrowed research topic and current project status are reflected in an updated project plan. The updated plan maintains a realistic scope appropriate for the remaining project period while emphasizing design-led development, early prototyping, and user-centred evaluation.

The second half of the project will focus on:

- Completing the information architecture and requirements for the prototype fitness application.
- Creating a mobile application prototype with essential functionality that is simple yet useful.

- Using task-based techniques and cognitive walkthroughs to test usability with peers.
- Analysing evaluation data to assess how effectively the prototype supports coherent user journeys.
- Evaluating results critically and seeking areas for improvement and further research.

The updated Gantt chart now more clearly distinguishes the stages of research consolidation, prototype development, and evaluation. Time has been set aside expressly for iteration and reflection in order to account for unforeseen challenges and ensure that assessment activities are carried out rigorously rather than as a last-minute add-on.



In order to ensure that the remaining work can be finished to a high grade within the allotted time, these modifications are meant to enhance project deliverability and analytical depth.

4. Planned Technical Direction

The technical environment and tools needed to construct the fitness application prototype have been taken into consideration as the project moves into the development phase. Choosing a development strategy that facilitates quick prototyping, usability testing, and iterative improvement over full-scale commercial deployment is the main goal at this point.

Many mobile programming frameworks have been considered, such as native Android development with Android Studio and cross-platform choices like Flutter. Each option was evaluated according to its familiarity with development, compliance with the anticipated feature set, testability, and adherence to project constraints.

Currently, the best option is to use Android Studio for native Android programming. This choice is influenced by the Android ecosystem's suitability for fitness and health-related apps, the accessibility of testing and development resources, and previous development experience. Android Studio's direct access to platform-native components enables precise control over navigation patterns, interaction behaviour, and performance while evaluating user journeys and cognitive strain.

Although cross-platform frameworks were taken into consideration due to their potential efficiency and wider device reach, the project's scope and academic focus made these benefits less important. The project does not need to be deployed across several operating systems, and giving platform familiarity top priority lowers technical risk and frees up more time for design quality assessment.

Development will now concentrate on creating a simple yet useful prototype that complements the main user journeys. This will contain rudimentary data persistence as needed, basic feature navigation, and placeholder visualization to aid in usability testing. In order to keep the project's scope modest and guarantee that evaluation activities remain at its core, advanced functionality, optimization, and deployment considerations are purposefully left out.

Rather than showcasing technical intricacy, this technical approach advances the project's main goal of examining user experiences. The project is positioned to produce a prototype that is appropriate for significant user-centred evaluation in the second half of the project by using technologies that minimize overhead and maximize control.

Conclusion

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5. Conclusion

The project's present status has been described in this interim report, along with the advancements made during the first half of the development period and the intended course for the remaining work. Through background research on multipurpose mobile applications and fragmented user journeys, a distinct project focus has been formed, offering a solid conceptual framework for the subsequent design and development stages.

Consolidating the literature review, analysing current application paradigms, and adjusting the project's scope in response to research findings have all advanced significantly. Although some elements of the initial project plan have been modified, these modifications are more indicative of a better comprehension of the issue area than of delays or failures. The revised project plan and Gantt chart show a methodical and practical strategy to finishing the remaining tasks within the allotted time.

The project's next stage will go from research-driven investigation to real-world application. A technically viable development approach has been determined, and the focus will be on creating a working prototype that can facilitate significant usability assessment. To evaluate how well the prototype tackles the issues highlighted in the literature, user testing and critical reflection will be essential.

All things considered, the project is moving forward as planned and is in a strong position to achieve its stated goals. In order to ensure that the final project provides a cogent, well-supported examination into multifunctional mobile app design within the fitness domain, feedback received at this interim stage will be used to improve both the technical and analytical aspects of the work.